

ETHICS & ISSUES 1-2

What Should You Share on the Internet?

The Internet was born in the waning days of the 1960s. Many of the early users of the Internet grew up during that time, and their connection with the era's vision of peace, harmony, and sharing quickly became a defining characteristic of the early Internet. Often, the Internet is equated with a public square, where ideas and experiences are shared freely. As the Internet has become a mainstream, or even necessary, tool, the easygoing ways of those days have been reeled in somewhat. While the Internet now makes the comparison to a public square a reality on a grand scale, laws and good taste require otherwise, and some people continue to make bad decisions in this regard. For example, if you believe that sharing your experience of a good song is the right thing to do, the law says you may not copy that song and provide it to others on the Internet. Other people sometimes share inappropriate or embarrassing videos on the Internet, leading to personal problems and sometimes lawsuits. Some people videotape newly released movies in a movie theater and then share that video with nonpaying viewers on the Internet. Camera-enabled cell phones allow people to capture and send pictures and videos at a moment's notice. Would your conscience or fear of the law stop you from copying a favorite song for a friend over the Internet? Why? If you see a person secretly video-recording a movie in a theater, would you report the incident? Why or why not? How can businesses and governments protect privacy with so many digital cameras and camera-enabled cell phones in the world?

COMPUTER SOFTWARE

Software, also called a **program**, consists of a series of instructions that tells the computer what to do and how to do it.

You interact with a program through its user interface. The user interface controls how you enter data and instructions and how information is displayed on the screen. Software today often has a graphical user interface. With a **graphical user interface (GUI pronounced gooey)**, you interact with the software

using text, graphics, and visual images such as icons. An **icon** is a miniature image that represents a program, an instruction, or some other object. You can use the mouse to select icons that perform operations such as starting a program.

The two categories of software are system software and application software.

System Software

System software consists of the programs that control or maintain the operations of the computer and its devices. System software serves as the interface between the user, the application software, and the computer's hardware. Two types of system software are the operating system and utility programs.

OPERATING SYSTEM An operating system is a set of programs that coordinates all the activities among computer hardware devices. It provides a means for users to communicate with the computer and other software. Many of today's computers use Windows Vista (Figure 1-10) or Windows XP, which are two of Microsoft's operating systems, or Mac OS X, Apple's operating system.

WEB LINK 1-6**Windows Vista**

For more information, visit scs.site.com/dc2009/ch1/weblink and then click Windows Vista.

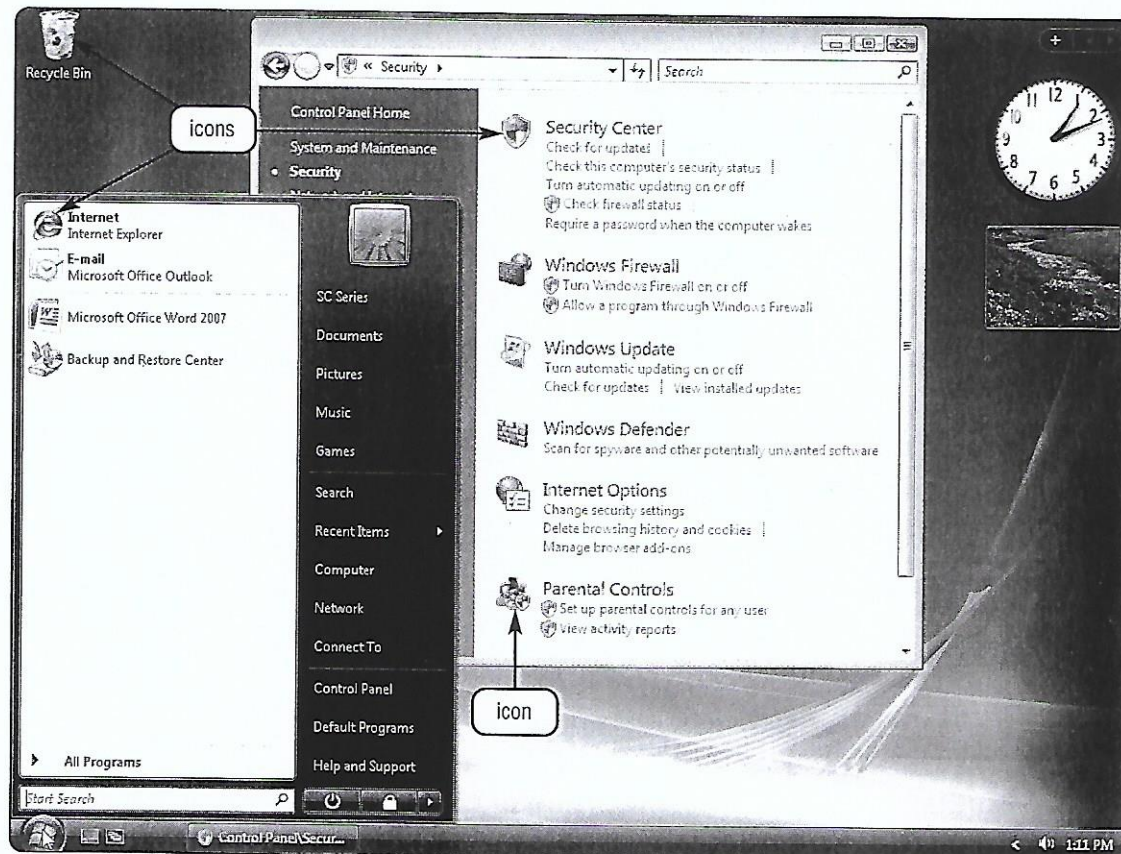


FIGURE 1-10 The graphical user interface of Windows Vista, Microsoft's latest operating system.

When a user starts a computer, portions of the operating system load into memory from the computer's hard disk. It remains in memory while the computer is on.

UTILITY PROGRAM A *utility program* allows a user to perform maintenance-type tasks usually related to managing a computer, its devices, or its programs. For example, you can use a utility program to transfer digital photos to a CD or DVD. Most operating systems include several utility programs for managing disk drives, printers, and other devices and media. You also can buy utility programs that allow you to perform additional computer management functions.

Application Software

Application software consists of programs designed to make users more productive and/or assist them with personal tasks. A widely used type of application software related to communications is a Web browser, which allows users with an Internet connection to access and view Web pages. Other popular application software includes word processing software, spreadsheet software, database software, and presentation graphics software.

Many other types of application software exist that enable users to perform a variety of tasks. These include personal information management, note taking, project management,

accounting, document management, computer-aided design, desktop publishing, paint/image editing, photo editing, audio and video editing, multimedia authoring, Web page authoring, personal finance, legal, tax preparation, home design/landscaping, travel and mapping, education, reference, and entertainment (e.g., games or simulations, etc.). As shown in Figure 1-11, you can purchase application software from a store that sells computer products. Read Ethics & Issues 1-3 for a related discussion.



FIGURE 1-11 Stores that sell computer products have shelves stocked with software for sale.

ETHICS & ISSUES 1-3

What Can Be Done to Combat Computer Addiction?

The notion that a person can become addicted to computer use has existed since the 1970s. Today, the extensive list of activities that people engage in with computers provides an ever growing list of reasons why some become computer addicted. While experts disagree on the specific definition of computer addiction, they generally agree that a person is computer addicted when obsessive use of a computer negatively impacts health, personal life, and professional life. The media reports extreme examples of computer addiction with growing frequency. One man died after playing a computer game for 86 hours while foregoing food and sleep. Another died after spending almost a week using a computer. Such incidents are rare, but experts consider the problem to be widespread, especially among children. China classifies more than 2.6 million people under the age of 18 as computer addicts. Sometimes mistakenly, parents feel that computer use is a healthier alternative to television or video games and, therefore, permit excessive computer use. For adults, computer addiction often leads to less time spent with family and has proven to be as potent a cause for divorce as gambling or substance abuse. What steps should parents and society take to combat computer addiction? Why? How can software providers help to alleviate the problem? Should certain software include built-in timers to disable the software after a certain number of hours, and then not enable it again until a set number of hours has elapsed? Why or why not? Do you believe that those identified as computer addicts should receive health insurance benefits for counseling services? Why or why not?